

# Integral Transforms and Fractional Calculus for the Fabius and Rvachev Functions

Exact antiderivative hierarchies, inverse-function transmutation, Mellin–Laplace bridges, and a Gamma–zeta fractional-dilation defect

Frontier research report based on the recursive `*.tex` corpus under  
`ProveIt/Analysis/FabiusFunction/docs`

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## Abstract

Let  $F$  be the bounded Fabius distribution function, let  $\mathcal{F}$  be the signed Thue–Morse continuation, and let  $G = F^{-1}$  on  $[0, 1]$ . This report audits the recursive LaTeX documentation tree of the `ProveIt` Fabius project and develops an integral and fractional-calculus layer that is not systematically present in the audited corpus. The novelty claims are deliberately relative to that corpus, not claims of exhaustive worldwide priority.

The ordinary dilation equation yields complete  $n$ -fold primitive and derivative ladders. For example,

$$I_{0+}^n F(x) = 2^{\binom{n}{2}} F(x/2^n), \quad 0 \leq x \leq 1,$$

and the same identity is global for  $\mathcal{F}$ . Polynomial and analytic weights are handled by finite or convergent operator expansions. A nonlinear change-of-variables theorem converts fractional integrals of  $h(F)$  into integrals involving the inverse  $G$ , and conversely. It includes the explicit primitive

$$\int G(y) \, dy = yG(y) - F\left(\frac{G(y)}{2}\right) + C$$

and exact formulas for fractional integrals and derivatives of  $G$  that avoid differentiating its singular endpoint asymptotics.

The probability model  $Y = \sum_{j \geq 1} 2^{-j} V_j$ ,  $V_j \sim \text{Unif}[0, 1]$ , and  $Z = 2Y - 1$  organizes the definite integrals. Its complex moment  $\mu(s) = \mathbb{E}(Y^s)$  gives Mellin transforms, repeated-primitive endpoint values, logarithmic moments, and the exact fractional dyadic lattice

$$(I_{0+}^\beta F)(2^{-n}) = 2^{-n\beta - \binom{n}{2}} \frac{\mu(n + \beta)}{\Gamma(n + \beta + 1)}.$$

The integer moments follow from Bernoulli cumulants and complete Bell polynomials. Laplace, Fourier, Mellin, Stieltjes, Cauchy, autocorrelation, and fractional Sobolev transforms are developed in a common normalization.

The main frontier theorem introduces the continuous dyadic Appell family

$$(\mathfrak{A}_q \mathcal{F})(x) = 2^{q(q-1)/2} \mathcal{F}(2^{-q}x), \quad q \in \mathbb{R}.$$

It interpolates all integer primitives and derivatives but is not the Riemann–Liouville semigroup at noninteger order. Its exact Laplace-space defect is

$$\frac{\mathcal{L}\{\mathfrak{A}_q \mathcal{F}\}(s)}{s^{-q} \mathcal{L}\{\mathcal{F}\}(s)} = \exp(P(\log_2 s + q) - P(\log_2 s)),$$

where  $P$  is the same one-periodic Gamma–zeta correction that occurs in the audited Lambert- $W$  endpoint theory. Since every nonzero Fourier coefficient of its centered part is nonzero, the defect vanishes identically exactly for  $q \in \mathbb{Z}$ . Thus integer calculus is the rigid zero-defect sublattice of a natural continuous dyadic interpolation.

Finally, the endpoint quadratic law implies a Gevrey-2, zero-radius Taylor barrier for the Mellin transform of the inverse, and moment expansions give convergent exterior series for the Riesz fractional Laplacian and potential of up. All numerical evidence is generated by the commented Python program distributed with the report.

**Proof-status convention.** Statements marked **proved here** are proved in this report from definitions and standard analysis. Statements marked **derived from audited inputs** combine such arguments with a theorem already present in the audited repository, chiefly the exact periodic negative-Laplace decomposition or the endpoint quadratic law. Decimal values and plots are **numerically checked**. Conjectures are marked **conjectural/open**. None of the new results is represented as already formalized in Lean.

## Contents

|          |                                                                   |           |
|----------|-------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Corpus audit, scope, and novelty discipline</b>                | <b>5</b>  |
| 1.1      | Recursive source boundary . . . . .                               | 5         |
| 1.2      | Headline additions relative to the audited tree . . . . .         | 5         |
| <b>2</b> | <b>Four equivalent languages and normalization</b>                | <b>6</b>  |
| 2.1      | The three functions . . . . .                                     | 6         |
| 2.2      | Probability model and moments . . . . .                           | 6         |
| 2.3      | Integral conventions . . . . .                                    | 7         |
| <b>3</b> | <b>Ordinary primitives and the full integer-order ladder</b>      | <b>7</b>  |
| 3.1      | The first primitive and its totalization . . . . .                | 7         |
| 3.2      | All integer primitives . . . . .                                  | 8         |
| 3.3      | Integer derivatives . . . . .                                     | 9         |
| 3.4      | The up-function primitive hierarchy . . . . .                     | 9         |
| <b>4</b> | <b>Weighted antiderivatives and analytic weights</b>              | <b>10</b> |
| 4.1      | Monomial weights: a finite closed form . . . . .                  | 10        |
| 4.2      | Complex powers and dyadic-shell series . . . . .                  | 10        |
| 4.3      | Truncated moment–quantile formula . . . . .                       | 11        |
| 4.4      | Analytic weights . . . . .                                        | 11        |
| 4.5      | Weights for up . . . . .                                          | 12        |
| <b>5</b> | <b>Nonlinear transmutation between a function and its inverse</b> | <b>12</b> |

|           |                                                                          |           |
|-----------|--------------------------------------------------------------------------|-----------|
| 5.1       | A fractional change-of-variables theorem . . . . .                       | 12        |
| 5.2       | Ordinary inverse antiderivatives . . . . .                               | 13        |
| 5.3       | Mixed moments of $F$ and $G$ . . . . .                                   | 13        |
| 5.4       | Ordinary and fractional derivatives of $G$ . . . . .                     | 14        |
| 5.5       | Sharp $L^r$ threshold for the inverse slope . . . . .                    | 14        |
| <b>6</b>  | <b>Definite integrals, moments, and Bell–Bernoulli structure</b>         | <b>15</b> |
| 6.1       | Mellin transforms of $F$ and $\text{up}$ . . . . .                       | 15        |
| 6.2       | Bernoulli cumulants and complete Bell moments . . . . .                  | 16        |
| 6.3       | Bernoulli-polynomial integrals . . . . .                                 | 16        |
| 6.4       | Moments of ordinary derivatives . . . . .                                | 17        |
| 6.5       | Beta integrals and products . . . . .                                    | 17        |
| 6.6       | A mixed product and a sinc-product integral . . . . .                    | 18        |
| <b>7</b>  | <b>Laplace, Fourier, Mellin, Cauchy, and correlation transforms</b>      | <b>18</b> |
| 7.1       | Laplace transforms of $F$ and $\text{up}$ . . . . .                      | 18        |
| 7.2       | Laplace and Mellin transforms of the inverse . . . . .                   | 19        |
| 7.3       | Stieltjes and Cauchy transforms . . . . .                                | 20        |
| 7.4       | Autocorrelation, convolution, and Sobolev energies . . . . .             | 20        |
| <b>8</b>  | <b>Riemann–Liouville and Caputo calculus on dyadic scales</b>            | <b>21</b> |
| 8.1       | The exact order-shift equation . . . . .                                 | 21        |
| 8.2       | Endpoint values and the fractional dyadic lattice . . . . .              | 21        |
| 8.3       | Fractional calculus for $\text{up}$ . . . . .                            | 22        |
| 8.4       | Fractional derivatives of $F$ . . . . .                                  | 22        |
| 8.5       | Complements and right-sided fractional integrals . . . . .               | 23        |
| <b>9</b>  | <b>A continuous dyadic Appell family and its exact fractional defect</b> | <b>23</b> |
| 9.1       | The continuous-order dyadic family . . . . .                             | 23        |
| 9.2       | Laplace transform of the signed continuation . . . . .                   | 24        |
| 9.3       | Exact completion of the periodic endpoint term . . . . .                 | 24        |
| 9.4       | The exact Riemann–Liouville defect . . . . .                             | 25        |
| 9.5       | Interpretation . . . . .                                                 | 27        |
| <b>10</b> | <b>Whole-line fractional operators acting on <math>\text{up}</math></b>  | <b>27</b> |
| 10.1      | Fractional Laplacian outside the support . . . . .                       | 28        |

|                                                                                  |           |
|----------------------------------------------------------------------------------|-----------|
| 10.2 Riesz potentials . . . . .                                                  | 28        |
| 10.3 Transform-side fractional derivatives . . . . .                             | 28        |
| <b>11 The inverse Mellin transform has a Gevrey-2 boundary</b>                   | <b>29</b> |
| <b>12 Numerical experiments and reproducibility</b>                              | <b>30</b> |
| 12.1 Numerical cautions . . . . .                                                | 30        |
| <b>13 Conjectures and research directions</b>                                    | <b>31</b> |
| 13.1 Higher inverse derivatives . . . . .                                        | 31        |
| 13.2 Fractional endpoint transseries . . . . .                                   | 31        |
| 13.3 Physical-space sign structure of the dyadic defect . . . . .                | 31        |
| 13.4 A Jackson or $q$ -fractional calculus at base $1/2$ . . . . .               | 31        |
| 13.5 Arithmetic and closed forms for Sobolev energies . . . . .                  | 32        |
| 13.6 Nonlinear fractional Volterra equations for $G$ . . . . .                   | 32        |
| 13.7 Complex order and monodromy . . . . .                                       | 32        |
| 13.8 Formalization program . . . . .                                             | 32        |
| <b>14 Summary of exact formulas</b>                                              | <b>33</b> |
| <b>A Detailed proof supplements</b>                                              | <b>34</b> |
| A.1 Entire continuation in power exponents . . . . .                             | 34        |
| A.2 Moment recurrence . . . . .                                                  | 34        |
| A.3 Why the inverse Mellin integral stops at $\operatorname{Re} s = 0$ . . . . . | 34        |
| A.4 A compact derivation of the Gamma–zeta defect . . . . .                      | 34        |
| <b>B Numerical files and execution</b>                                           | <b>35</b> |
| <b>C Audit notes and source map</b>                                              | <b>35</b> |

# 1 Corpus audit, scope, and novelty discipline

## 1.1 Recursive source boundary

The source boundary was the public recursive tree under `ProveIt/Analysis/FabiusFunction/docs` as accessed on 27 August 2026. The tree contains living synthesis documents, a consolidated non-formalized-frontier volume, active frontier drafts, historical snapshots, and corrected or transcribed source papers. A contemporaneous recursive inventory in the tree records 67 distinct `.tex` paths: five living synthesis/reference documents, thirty-two frozen historical sources, twenty-three active frontier drafts (twelve of them Fourier-decay dossiers), and seven transcribed or corrected papers. Historical copies were read as lineages: repetition across a frozen draft and its successor does not count as independent prior art.

The audit treated as established repository material, and therefore did not claim as new:

- the separation of the bounded  $F$ , the compactly supported up, and the signed global continuation  $\mathcal{F}$ ;
- exact dyadic values, moment formulas, Thue–Morse finite sums, half-base  $q$ -Pochhammer and Gaussian-binomial identities;
- repeated-integration splines and numerical integration experiments already occurring in the broad frontier notebooks;
- the infinite sinc product, its integer zeros, spectral zeta/determinant material, half-integer identities, Fourier-decay dossiers, and Pascal-weighted generalizations;
- the exact negative-Laplace decomposition, its nonconstant log-periodic Gamma–zeta term, the lower Lambert- $W_{-1}$  saddle, all-order Bell-polynomial expansions, and inverse endpoint asymptotics;
- Lagrange and Richardson extrapolation systems already developed in the repository.

Targeted searches were then made for `fractional calculus`, `Riemann-Liouville`, `Caputo`, general fractional antiderivatives, inverse fractional transmutation, and formulas equivalent to the continuous-order dyadic defect below. No systematic counterpart was found in the living/current lineage. The present report is therefore designed to be complementary: old inputs are recalled briefly, while proofs and frontier claims concentrate on integral calculus.

## 1.2 Headline additions relative to the audited tree

The following package appears to be new relative to the audited tree:

1. a complete ordinary and fractional weighted-antiderivative calculus based on the kernels  $I_{0+}^\alpha F$  and  $I_{-1+}^\alpha$  up;
2. nonlinear Riemann–Liouville transmutation identities between  $F$  and  $G = F^{-1}$ ;
3. exact fractional dyadic values in terms of the analytic moment  $\mu(n + \beta)$ ;
4. a continuous dyadic Appell interpolation of all integer derivatives and primitives;
5. the exact Gamma–zeta multiplier measuring its failure to equal the Riemann–Liouville semigroup, together with integer-order rigidity;
6. exterior moment series for the Riesz fractional Laplacian and Riesz potential of up;

7. a Gevrey-2, zero-radius Taylor theorem for the inverse Mellin transform at its natural boundary;
8. an inverse-derivative  $L^r$  threshold and several mixed  $F$ – $G$  integral bridges.

The word “new” below always means new relative to the audited repository and apparently nonstandard. A worldwide priority assertion would require a separate bibliographic search far beyond the source-comparison purpose of this report.

## 2 Four equivalent languages and normalization

### 2.1 The three functions

**Definition 2.1** (Bounded Fabius function). Following the classical and project normalizations [1, 3, 12], the bounded Fabius function is the  $C^\infty$  map  $F : \mathbb{R} \rightarrow [0, 1]$  satisfying

$$F(x) = 0 \quad (x \leq 0), \quad F(x) = 1 \quad (x \geq 1), \quad (1)$$

$$F(1 - x) = 1 - F(x) \quad (0 \leq x \leq 1), \quad (2)$$

$$F'(x) = 2F(2x) \quad (0 \leq x \leq \tfrac{1}{2}). \quad (3)$$

Its inverse on  $[0, 1]$  is denoted by

$$G = F^{-1} : [0, 1] \longrightarrow [0, 1].$$

**Definition 2.2** (Rvachev up-function). The compactly supported even bump is

$$\text{up}(x) = F(1 - |x|) = \begin{cases} F(x + 1), & x \leq 0, \\ F(1 - x), & x > 0. \end{cases} \quad (4)$$

Thus  $\text{supp up} = [-1, 1]$ ,  $\text{up}(0) = 1$ , and

$$F(x) = \text{up}(x - 1), \quad 0 \leq x \leq 1. \quad (5)$$

**Definition 2.3** (Signed Thue–Morse continuation). Let  $\varepsilon_n = (-1)^{s_2(n)}$ , where  $s_2(n)$  is the binary digit sum. Define

$$\mathcal{F}(x) = \sum_{n \geq 0} \varepsilon_n \text{up}(x - 2n - 1). \quad (6)$$

The sum is locally finite,  $\mathcal{F} = F$  on  $(-\infty, 1]$ , and the audited global equation is

$$\mathcal{F}'(x) = 2\mathcal{F}(2x), \quad x \in \mathbb{R}. \quad (7)$$

The bounded and signed objects must not be interchanged to the right of 1. The global object is ideal for pure dilation calculus; the bounded object is ideal for probability, Mellin transforms, and inverse-function questions.

### 2.2 Probability model and moments

Let  $V_1, V_2, \dots$  be independent uniform random variables on  $[0, 1]$  and set

$$Y = \sum_{j=1}^{\infty} 2^{-j} V_j, \quad Z = 2Y - 1. \quad (8)$$

Then  $F$  is the CDF of  $Y$ , while  $\text{up}$  is the density of  $Z$ ; this random-series model is classical in the subject [4, 5, 13]. Consequently

$$F'(x) = 2 \text{up}(2x - 1), \quad \int_{-1}^1 \text{up}(x) \, dx = 1. \quad (9)$$

We use

$$\mu(s) = \mathbb{E}(Y^s) = \int_0^1 x^s F'(x) \, dx, \quad m_n = \mathbb{E}(Z^n) = \int_{-1}^1 x^n \text{up}(x) \, dx. \quad (10)$$

Because  $F'$  is flat at both endpoints,  $\mu(s)$  is entire in  $s$ . The symmetry  $Y \stackrel{d}{=} 1 - Y$  and  $Z \stackrel{d}{=} -Z$  will repeatedly remove odd terms.

### 2.3 Integral conventions

We use the standard left-sided Riemann–Liouville and Caputo conventions [7, 8, 10]. For  $a \in \mathbb{R}$  and  $\text{Re } \alpha > 0$ , the left Riemann–Liouville integral is

$$(I_{a+}^\alpha f)(x) = \frac{1}{\Gamma(\alpha)} \int_a^x (x-t)^{\alpha-1} f(t) \, dt. \quad (11)$$

We abbreviate

$$\mathcal{J}_\alpha(x) = (I_{0+}^\alpha F)(x), \quad \mathcal{K}_\alpha(x) = (I_{-1+}^\alpha \text{up})(x). \quad (12)$$

The order-zero convention is  $I^0 f = f$ . For  $n-1 < \alpha < n$ , the left Riemann–Liouville and Caputo derivatives are

$$D_{a+}^\alpha f = \frac{d^n}{dx^n} I_{a+}^{n-\alpha} f, \quad (13)$$

$${}^C D_{a+}^\alpha f = I_{a+}^{n-\alpha} f^{(n)}. \quad (14)$$

If  $f^{(k)}(a) = 0$  for  $0 \leq k < n$ , they coincide. This applies to  $F$  at 0 and to  $\text{up}$  at  $-1$  for every finite order.

Our Fourier convention is

$$\widehat{f}(t) = \int_{\mathbb{R}} f(x) e^{-itx} \, dx, \quad f(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{f}(t) e^{itx} \, dt. \quad (15)$$

Here  $\text{sinc } z = \sin z/z$  with  $\text{sinc } 0 = 1$ .

## 3 Ordinary primitives and the full integer-order ladder

### 3.1 The first primitive and its totalization

**Theorem 3.1** (First primitive of the bounded function). *For every real  $x$ ,*

$$\int_0^x F(t) \, dt = \begin{cases} 0, & x \leq 0, \\ F(x/2), & 0 \leq x \leq 1, \\ x - \frac{1}{2}, & x \geq 1. \end{cases} \quad (16)$$

*The middle formula is [proved here](#).*

*Proof.* For  $0 \leq x \leq 1$ , differentiation and equation (3) give

$$\frac{d}{dx}F(x/2) = \frac{1}{2}F'(x/2) = F(x),$$

and both sides vanish at  $x = 0$ . Symmetry gives  $\int_0^1 F = 1/2$ . Since  $F = 1$  to the right of 1, the final branch is  $1/2 + x - 1$ .  $\square$

**Corollary 3.2** (Complement and reflected primitives). *For  $0 \leq x \leq 1$ ,*

$$\int_0^x (1 - F(t)) dt = x - F(x/2), \quad (17)$$

$$\int_0^x F(1 - t) dt = x - F(x/2), \quad (18)$$

$$\int_0^x \text{up}(t) dt = x - F(x/2). \quad (19)$$

### 3.2 All integer primitives

**Theorem 3.3** (Dyadic primitive hierarchy). *For every integer  $n \geq 0$ ,*

$$(I_{0+}^n F)(x) = 2^{\binom{n}{2}} F(x/2^n), \quad 0 \leq x \leq 1. \quad (20)$$

*For the signed continuation, the same formula holds globally:*

$$(I_{0+}^n \mathcal{F})(x) = 2^{\binom{n}{2}} \mathcal{F}(x/2^n), \quad x \in \mathbb{R}, \quad (21)$$

*where the anchored iterated integral is understood for  $x < 0$  with its usual orientation. Both statements are **proved here**.*

*Proof.* The case  $n = 0$  is tautological and  $n = 1$  is theorem 3.1. If  $H_n(x) = 2^{\binom{n}{2}} F(x/2^n)$ , then, while  $0 \leq x \leq 1$ ,

$$H'_n(x) = 2^{\binom{n}{2}-n} F'(x/2^n) = 2^{\binom{n}{2}-n+1} F(x/2^{n-1}) = H_{n-1}(x).$$

The endpoint value is  $H_n(0) = 0$ . For  $\mathcal{F}$ , the same calculation uses the global dilation equation equation (7) without a domain restriction.  $\square$

**Corollary 3.4** (Endpoint primitive values and dyadic values). *For  $n \geq 1$ ,*

$$(I_{0+}^n F)(1) = \frac{\mu(n)}{n!} = 2^{\binom{n}{2}} F(2^{-n}). \quad (22)$$

*Thus*

$$F(2^{-n}) = 2^{-\binom{n}{2}} \frac{\mu(n)}{n!}. \quad (23)$$

*Proof.* The kernel formula gives

$$(I_{0+}^n F)(1) = \frac{1}{(n-1)!} \int_0^1 (1-t)^{n-1} F(t) dt.$$

By symmetry and integration by parts this equals  $\mu(n)/n!$ . The second equality is equation (20) at  $x = 1$ .  $\square$

### 3.3 Integer derivatives

**Theorem 3.5** (Derivative dilation ladder). *For every integer  $n \geq 0$ ,*

$$\mathcal{F}^{(n)}(x) = 2^{\binom{n+1}{2}} \mathcal{F}(2^n x), \quad x \in \mathbb{R}. \quad (24)$$

*Consequently, whenever  $0 \leq x \leq 2^{-n}$ ,*

$$F^{(n)}(x) = 2^{\binom{n+1}{2}} F(2^n x). \quad (25)$$

*This is an audited input, restated to pair it with theorem 3.3.*

**Corollary 3.6** (Local derivative moments). *For  $n \geq 0$  and  $\operatorname{Re} p > -1$ ,*

$$\int_0^{2^{-n}} x^p F^{(n)}(x) dx = 2^{\binom{n}{2} - np} \frac{1 - \mu(p+1)}{p+1}. \quad (26)$$

*The right side extends holomorphically in  $p$  after removing the apparent singularity.*

*Proof.* Use equation (25), put  $u = 2^n x$ , and use the Mellin identity proved in theorem 6.1 below.  $\square$

### 3.4 The up-function primitive hierarchy

**Theorem 3.7** (CDF and repeated primitives of up). *For every  $x \in \mathbb{R}$ ,*

$$\int_{-\infty}^x \operatorname{up}(t) dt = F\left(\frac{x+1}{2}\right). \quad (27)$$

*For each integer  $n \geq 1$ ,*

$$(I_{-1+}^n \operatorname{up})(x) = 2^{\binom{n}{2}} F\left(\frac{x+1}{2^n}\right), \quad -1 \leq x \leq 2^{n-1} - 1. \quad (28)$$

*In particular, it holds throughout the support interval  $[-1, 1]$ . These identities are [proved here](#).*

*Proof.* Differentiate  $F((x+1)/2)$  and use the unified identity  $F'(u) = 2 \operatorname{up}(2u - 1)$ . The function vanishes at  $x = -1$ . Repeated differentiation of the right side in equation (28) gives the preceding order whenever  $(x+1)/2^n \leq 1/2$ ; the base case is equation (27).  $\square$

**Proposition 3.8** (Polynomial continuation beyond the support). *For  $n \geq 1$  and  $x \geq 1$ ,*

$$(I_{-1+}^n \operatorname{up})(x) = \frac{1}{(n-1)!} \sum_{j=0}^{\lfloor (n-1)/2 \rfloor} \binom{n-1}{2j} m_{2j} x^{n-1-2j}. \quad (29)$$

*Thus every repeated primitive of the compact bump becomes an explicit moment polynomial once the integration point has passed the support. The statement is [proved here](#).*

*Proof.* For  $x \geq 1$  the kernel representation is

$$(I_{-1+}^n \operatorname{up})(x) = \frac{1}{(n-1)!} \int_{-1}^1 (x-t)^{n-1} \operatorname{up}(t) dt.$$

Expand the polynomial and use  $m_{2j+1} = 0$ .  $\square$

## 4 Weighted antiderivatives and analytic weights

### 4.1 Monomial weights: a finite closed form

**Theorem 4.1** (Weighted Riemann–Liouville expansion). *Let  $p \in \mathbb{N}_0$  and  $\operatorname{Re} \alpha > 0$ . Then*

$$I_{0+}^\alpha [t^p F(t)](x) = \sum_{k=0}^p (-1)^k \binom{p}{k} (\alpha)_k x^{p-k} \mathcal{J}_{\alpha+k}(x), \quad (30)$$

where  $(\alpha)_k = \Gamma(\alpha + k)/\Gamma(\alpha)$ . This is [proved here](#).

*Proof.* Inside the defining integral, expand

$$t^p = (x - (x - t))^p = \sum_{k=0}^p (-1)^k \binom{p}{k} x^{p-k} (x - t)^k.$$

The  $k$ th term changes the kernel order from  $\alpha$  to  $\alpha + k$  and contributes  $\Gamma(\alpha + k)/\Gamma(\alpha) = (\alpha)_k$ .  $\square$

**Corollary 4.2** (Closed  $n$ th antiderivative of  $x^p F(x)$ ). *For integers  $n \geq 1$ ,  $p \geq 0$ , and  $0 \leq x \leq 1$ ,*

$$I_{0+}^n [t^p F(t)](x) = \sum_{k=0}^p (-1)^k \binom{p}{k} (n)_k 2^{\binom{n+k}{2}} x^{p-k} F(x/2^{n+k}). \quad (31)$$

For  $n = 1$ , an ordinary indefinite antiderivative is

$$\boxed{\int x^p F(x) \, dx = \sum_{k=0}^p (-1)^k \frac{p!}{(p-k)!} 2^{\binom{k+1}{2}} x^{p-k} F(x/2^{k+1}) + C} \quad (32)$$

throughout  $0 \leq x \leq 1$ . Replacing  $F$  by  $\mathcal{F}$  makes both formulas global.

**Example 4.3.** The first cases are

$$\begin{aligned} \int F(x) \, dx &= F(x/2) + C, \\ \int xF(x) \, dx &= xF(x/2) - 2F(x/4) + C, \\ \int x^2 F(x) \, dx &= x^2 F(x/2) - 4xF(x/4) + 16F(x/8) + C, \\ \int x^3 F(x) \, dx &= x^3 F(x/2) - 6x^2 F(x/4) + 48xF(x/8) - 384F(x/16) + C. \end{aligned}$$

The rapidly growing coefficients are offset by the extraordinary flatness of the dyadically contracted  $F$  terms.

### 4.2 Complex powers and dyadic-shell series

For noninteger powers, a finite elementary combination is not expected, but two exact representations remain available.

**Proposition 4.4** (Binomial fractional-weight series). *Let  $\operatorname{Re} \alpha > 0$ ,  $\operatorname{Re} p > -1$ , and  $x > 0$ . Then*

$$I_{0+}^{\alpha}[t^p F(t)](x) = \sum_{k=0}^{\infty} (-1)^k \binom{p}{k} (\alpha)_k x^{p-k} \mathcal{J}_{\alpha+k}(x). \quad (33)$$

*The series converges absolutely. It terminates when  $p \in \mathbb{N}_0$ .*

*Proof.* Use the binomial series  $t^p = x^p(1 - (x - t)/x)^p$  and dominate with  $0 \leq \mathcal{J}_{\alpha+k}(x) \leq x^{\alpha+k}/\Gamma(\alpha + k + 1)$ . The resulting scalar terms are  $\mathcal{O}(k^{-\operatorname{Re} p - 2})$ .  $\square$

**Proposition 4.5** (Dyadic-shell series for arbitrary powers). *For fixed  $x \in (0, 1]$  and every  $p \in \mathbb{C}$ ,*

$$\int_0^x t^p F(t) dt = x^{p+1} \sum_{n=0}^{\infty} 2^{-n(p+1)} \int_{1/2}^1 u^p F(x2^{-n}u) du. \quad (34)$$

*The series is absolutely and locally uniformly convergent in  $p$ ; hence the weighted primitive is entire in the exponent. This is [proved here](#).*

*Proof.* Partition  $(0, x]$  into the shells  $[x2^{-n-1}, x2^{-n}]$  and substitute  $t = x2^{-n}u$ . Endpoint flatness of  $F$  dominates every factor  $2^{n|\operatorname{Re} p|}$  on compact  $p$ -sets.  $\square$

### 4.3 Truncated moment–quantile formula

**Theorem 4.6** (Power-weighted primitive through the inverse). *For  $p \neq -1$  and  $0 < x \leq 1$ ,*

$$\int_0^x t^p F(t) dt = \frac{x^{p+1}F(x) - \int_0^{F(x)} G(u)^{p+1} du}{p+1}. \quad (35)$$

*At  $p = -1$  the removable logarithmic form is*

$$\int_0^x \frac{F(t)}{t} dt = F(x) \log x - \int_0^{F(x)} \log G(u) du. \quad (36)$$

*These formulas are [proved here](#) and extend holomorphically in  $p$ .*

*Proof.* Stieltjes integration by parts gives

$$\int_0^x t^p F(t) dt = \frac{x^{p+1}F(x) - \int_0^x t^{p+1} dF(t)}{p+1}.$$

The monotone substitution  $u = F(t)$  turns the Stieltjes integral into  $\int_0^{F(x)} G(u)^{p+1} du$ . The logarithmic identity follows by using  $\log t$  as the antiderivative of  $1/t$  and then the same substitution.  $\square$

### 4.4 Analytic weights

**Theorem 4.7** (Local analytic-weight expansion). *Suppose  $g$  is analytic in a disk centered at  $x > 0$  that contains  $[0, x]$ , and  $\operatorname{Re} \alpha > 0$ . Then*

$$I_{0+}^{\alpha}[g(t)F(t)](x) = \sum_{k=0}^{\infty} \frac{(-1)^k (\alpha)_k}{k!} g^{(k)}(x) \mathcal{J}_{\alpha+k}(x). \quad (37)$$

*The convergence is absolute under any Cauchy estimate with radius strictly larger than  $x$ .*

*Proof.* Insert the Taylor series  $g(t) = \sum_{k \geq 0} g^{(k)}(x)(t-x)^k/k!$  into equation (11) and integrate termwise.  $\square$

**Corollary 4.8** (Exponential and trigonometric weights). *For  $\lambda \in \mathbb{C}$ ,*

$$I_{0+}^{\alpha} [e^{\lambda t} F(t)](x) = e^{\lambda x} \sum_{k=0}^{\infty} \frac{(-\lambda)^k (\alpha)_k}{k!} \mathcal{J}_{\alpha+k}(x). \quad (38)$$

*Taking real or imaginary parts gives the corresponding cosine and sine formulas.*

## 4.5 Weights for up

The same finite algebra works after shifting the lower endpoint.

**Proposition 4.9** (Shifted weighted up-integrals). *For  $p \in \mathbb{N}_0$ ,  $\operatorname{Re} \alpha > 0$ , and  $x \geq -1$ ,*

$$I_{-1+}^{\alpha} [(t+1)^p \operatorname{up}(t)](x) = \sum_{k=0}^p (-1)^k \binom{p}{k} (\alpha)_k (x+1)^{p-k} \mathcal{K}_{\alpha+k}(x). \quad (39)$$

*Ordinary powers  $t^p$  follow by expanding  $t^p = ((t+1) - 1)^p$ .*

# 5 Nonlinear transmutation between a function and its inverse

## 5.1 A fractional change-of-variables theorem

**Theorem 5.1** (Fabius–quantile transmutation). *Let  $\operatorname{Re} \alpha > 0$ ,  $0 \leq x \leq 1$ , and  $h \in C^1([0, 1])$ . Then*

$$I_{0+}^{\alpha} [h(F)](x) = \frac{h(0)x^{\alpha}}{\Gamma(\alpha+1)} + \frac{1}{\Gamma(\alpha+1)} \int_0^{F(x)} h'(u)(x-G(u))^{\alpha} du. \quad (40)$$

*Dually, for  $0 \leq y \leq 1$ ,*

$$I_{0+}^{\alpha} [h(G)](y) = \frac{h(0)y^{\alpha}}{\Gamma(\alpha+1)} + \frac{1}{\Gamma(\alpha+1)} \int_0^{G(y)} h'(x)(y-F(x))^{\alpha} dx. \quad (41)$$

*Both formulas are [proved here](#).*

*Proof.* For a differentiable  $f$ ,

$$I_{0+}^{\alpha} f(x) = \frac{f(0)x^{\alpha}}{\Gamma(\alpha+1)} + \frac{1}{\Gamma(\alpha+1)} \int_0^x (x-t)^{\alpha} f'(t) dt.$$

Apply this to  $f = h \circ F$  and substitute  $u = F(t)$ , then to  $f = h \circ G$  and substitute  $x = G(t)$ , equivalently  $t = F(x)$ .  $\square$

**Corollary 5.2** (Fractional integrals of the inverse and its powers). *For  $\operatorname{Re} \alpha > 0$ ,*

$$I_{0+}^{\alpha} G(y) = \frac{1}{\Gamma(\alpha+1)} \int_0^{G(y)} (y-F(x))^{\alpha} dx, \quad (42)$$

$$I_{0+}^{\alpha} [G^q](y) = \frac{q}{\Gamma(\alpha+1)} \int_0^{G(y)} x^{q-1} (y-F(x))^{\alpha} dx, \quad \operatorname{Re} q > 0. \quad (43)$$

*At the right endpoint,*

$$I_{0+}^{\alpha} G(1) = \frac{1}{\Gamma(\alpha+1)} \int_0^1 F(x)^{\alpha} dx. \quad (44)$$

*Proof.* Set  $h(x) = x$  or  $x^q$  in equation (41). At  $y = 1$ , use symmetry to replace  $(1 - F(x))^\alpha$  by  $F(x)^\alpha$  under the integral.  $\square$

## 5.2 Ordinary inverse antiderivatives

**Theorem 5.3** (Elementary primitive of  $G$ ). *For  $0 \leq y \leq 1$ ,*

$$\int G(y) dy = yG(y) - F\left(\frac{G(y)}{2}\right) + C. \quad (45)$$

*In particular,*

$$\int_0^1 G(y) dy = \frac{1}{2}. \quad (46)$$

*The formula is [proved here](#).*

*Proof.* Take  $\alpha = 1$  in equation (42):

$$\int_0^y G(u) du = yG(y) - \int_0^{G(y)} F(x) dx.$$

The remaining integral is  $F(G(y)/2)$  by theorem 3.1. At  $y = 1$ , use  $G(1) = 1$  and  $F(1/2) = 1/2$ .  $\square$

**Theorem 5.4** (Power-weighted inverse primitive). *For  $p \neq -1$ ,*

$$\int y^p G(y) dy = \frac{y^{p+1}G(y) - \int_0^{G(y)} F(x)^{p+1} dx}{p+1} + C. \quad (47)$$

*At  $p = -1$ ,*

$$\int \frac{G(y)}{y} dy = G(y) \log y - \int_0^{G(y)} \log F(x) dx + C. \quad (48)$$

*Both identities are [proved here](#).*

*Proof.* Integrate by parts and use  $G'(y) dy = dx$  under  $y = F(x)$ .  $\square$

## 5.3 Mixed moments of $F$ and $G$

**Proposition 5.5** (Master mixed-moment bridge). *Whenever the integrals converge,*

$$\int_0^1 y^a G(y)^b dy = \int_0^1 F(x)^a x^b F'(x) dx. \quad (49)$$

*In particular,*

$$\int_0^1 G(y)^b dy = \mu(b), \quad (50)$$

*and, for  $\operatorname{Re} a > -1$  and  $\operatorname{Re} b > 0$ ,*

$$\int_0^1 y^a G(y)^b dy = \frac{1}{a+1} \left( 1 - b \int_0^1 x^{b-1} F(x)^{a+1} dx \right). \quad (51)$$

*These formulas are [proved here](#).*

*Proof.* Substitute  $y = F(x)$  in the left side. Formula equation (50) is the case  $a = 0$ . For equation (51), write  $F^a F' = (a+1)^{-1} (F^{a+1})'$  and integrate by parts.  $\square$

## 5.4 Ordinary and fractional derivatives of $G$

**Proposition 5.6** (Recursive inverse derivatives). *On  $(0, 1)$  define*

$$R_1(x) = \frac{1}{F'(x)}, \quad R_{n+1}(x) = \frac{R'_n(x)}{F'(x)}. \quad (52)$$

*Then*

$$G^{(n)}(y) = R_n(G(y)). \quad (53)$$

*For example,*

$$G'(y) = \frac{1}{F'(G(y))}, \quad (54)$$

$$G''(y) = -\frac{F''(G(y))}{F'(G(y))^3}, \quad (55)$$

$$G'''(y) = \frac{3F''(G(y))^2 - F'(G(y))F'''(G(y))}{F'(G(y))^5}. \quad (56)$$

*Proof.* Differentiate  $F(G(y)) = y$  and iterate the chain rule. The recurrence follows because  $\frac{d}{dy}R_n(G(y)) = R'_n(G(y))/F'(G(y))$ .  $\square$

**Theorem 5.7** (Fractional derivative of the inverse). *For  $0 < \alpha < 1$  and  $0 < y < 1$ , the Riemann–Liouville and Caputo derivatives agree and*

$$D_{0+}^\alpha G(y) = \frac{1}{\Gamma(1-\alpha)} \int_0^{G(y)} (y - F(x))^{-\alpha} dx. \quad (57)$$

*The value is finite for every interior  $y$  and diverges at both endpoints  $y \downarrow 0$  and  $y \uparrow 1$ . The identity is [proved here](#); the two endpoint statements are [derived from audited inputs](#) (using the audited endpoint quadratic law).*

*Proof.* Because  $G(0) = 0$  and  $G$  is absolutely continuous,  $D_{0+}^\alpha G = I_{0+}^{1-\alpha} G'$ . Substitute  $u = F(x)$ , for which  $G'(u) du = dx$ . Interior finiteness follows from the local behavior of a monotone  $C^1$  change of variables away from the endpoints. At  $y = 1$ , the integral contains  $(1 - F(x))^{-\alpha}$  near  $x = 1$ ; by symmetry and the endpoint quadratic law this is not integrable. At the opposite endpoint,

$$D_{0+}^\alpha G(y) \geq \frac{y^{-\alpha} G(y)}{\Gamma(1-\alpha)}.$$

The endpoint quadratic law implies  $F(x) = o(x^N)$  for every  $N > 0$ , hence  $G(y)/y^\alpha \rightarrow \infty$  as  $y \downarrow 0$ .  $\square$

## 5.5 Sharp $L^r$ threshold for the inverse slope

**Theorem 5.8** (Inverse-slope integrability). *For  $r > 0$ ,*

$$\int_0^1 G'(y)^r dy = \int_0^1 F'(x)^{1-r} dx = 2^{-r} \int_{-1}^1 \text{up}(t)^{1-r} dt. \quad (58)$$

*This integral is finite if and only if  $0 < r \leq 1$ . The identity is [proved here](#); the “only if” direction is [derived from audited inputs](#) (using the audited endpoint quadratic law).*

*Proof.* Use  $y = F(x)$  and  $G'(F(x)) = 1/F'(x)$ , then  $F'(x) = 2 \operatorname{up}(2x - 1)$ . If  $r \leq 1$ , the exponent  $1 - r$  is nonnegative, so boundedness and compact support suffice. If  $r > 1$ , near  $x = 0$  the endpoint law

$$-\log F(x) \sim \frac{(\log(1/x))^2}{2 \log 2}$$

implies that  $F'(x)^{1-r}$  grows faster than  $\exp(c(\log(1/x))^2)$  for some  $c > 0$ , and its integral diverges after  $x = e^{-u}$ .  $\square$

**Corollary 5.9** (Negative powers and entropy). *For  $q > -1$ ,*

$$\int_0^1 G'(y)^{-q} dy = 2^q \int_{-1}^1 \operatorname{up}(t)^{q+1} dt. \quad (59)$$

Moreover,

$$\int_0^1 \log G'(y) dy = - \int_0^1 F'(x) \log F'(x) dx. \quad (60)$$

Thus the mean logarithmic slope of the quantile is the differential entropy of the Fabius density.

## 6 Definite integrals, moments, and Bell–Bernoulli structure

### 6.1 Mellin transforms of $F$ and $\operatorname{up}$

**Theorem 6.1** (Mellin–moment bridge). *For  $\operatorname{Re} s > 0$ ,*

$$\int_0^1 x^{s-1} F(x) dx = \frac{1 - \mu(s)}{s}, \quad (61)$$

$$\int_0^1 x^{s-1} \operatorname{up}(x) dx = \frac{\mu(s)}{s}. \quad (62)$$

The first expression extends to an entire function of  $s$ . For  $\operatorname{Re} q > -1$ ,

$$\mathbb{E}|Z|^q = \frac{2\mu(q+1)}{q+1}. \quad (63)$$

All three identities are [proved here](#).

*Proof.* Integrate equation (61) by parts against  $x^s/s$ . On  $[0, 1]$ ,  $\operatorname{up}(x) = 1 - F(x)$ , which gives equation (62). Evenness gives the final formula.  $\square$

**Corollary 6.2** (Power-weighted definite integrals). *For  $\operatorname{Re} p > -1$ ,*

$$\int_0^1 x^p F(x) dx = \frac{1 - \mu(p+1)}{p+1}, \quad (64)$$

$$\int_0^1 x^p \operatorname{up}(x) dx = \frac{\mu(p+1)}{p+1}. \quad (65)$$

At integer  $p$ , the values are rational.

**Corollary 6.3** (Logarithmic Fabius moments). *For  $n \in \mathbb{N}_0$ ,*

$$\int_0^1 \frac{(\log x)^n}{x} F(x) dx = -\frac{\mu^{(n+1)}(0)}{n+1}. \quad (66)$$

*Proof.* Differentiate the entire continuation of  $(1 - \mu(s))/s$  at  $s = 0$ .  $\square$

## 6.2 Bernoulli cumulants and complete Bell moments

The moment generating function is

$$\mathcal{A}(t) = \mathbb{E}e^{tY} = \prod_{j=1}^{\infty} \frac{e^{t/2^j} - 1}{t/2^j}. \quad (67)$$

Its logarithm immediately exposes Bernoulli numbers.

**Proposition 6.4** (Cumulants). *Let  $\kappa_n$  be the cumulants of  $Y$ . Then*

$$\kappa_1 = \frac{1}{2}, \quad \kappa_n = \frac{B_n}{n(2^n - 1)} \quad (n \geq 2), \quad (68)$$

where  $B_n$  are Bernoulli numbers with  $B_2 = 1/6$ . Consequently

$$\mu_n = \mathcal{B}_n(\kappa_1, \dots, \kappa_n), \quad (69)$$

where  $\mathcal{B}_n$  is the complete exponential Bell polynomial. The cumulants of  $Z = 2Y - 1$  are  $2^n \kappa_n$  for  $n \geq 2$ , and its odd moments vanish.

*Proof.* For  $V \sim \text{Unif}[0, 1]$ ,

$$\log \mathbb{E}e^{tV} = \frac{t}{2} + \sum_{n \geq 2} \frac{B_n}{n} \frac{t^n}{n!}.$$

Scale by  $2^{-j}$ , sum over  $j$ , and use  $\sum_{j \geq 1} 2^{-jn} = 1/(2^n - 1)$ . The standard moment–cumulant relation gives equation (69).  $\square$

Table 1: Exact moments generated from Bernoulli cumulants.

| $n$ | $\mu_n = \mathbb{E}(Y^n)$      | $m_n = \mathbb{E}(Z^n)$        |
|-----|--------------------------------|--------------------------------|
| 0   | 1                              | 1                              |
| 1   | $\frac{1}{2}$                  | 0                              |
| 2   | $\frac{5}{18}$                 | $\frac{1}{9}$                  |
| 3   | $\frac{1}{6}$                  | 0                              |
| 4   | $\frac{143}{1350}$             | $\frac{19}{675}$               |
| 5   | $\frac{19}{270}$               | 0                              |
| 6   | $\frac{1153}{23814}$           | $\frac{583}{59535}$            |
| 7   | $\frac{583}{17010}$            | 0                              |
| 8   | $\frac{1616353}{65063250}$     | $\frac{132809}{32531625}$      |
| 9   | $\frac{132809}{7229250}$       | 0                              |
| 10  | $\frac{134926369}{9762090030}$ | $\frac{46840699}{24405225075}$ |

## 6.3 Bernoulli-polynomial integrals

**Proposition 6.5** (Bernoulli-polynomial bridge). *Let  $B_n(x)$  be the Bernoulli polynomial. For  $n \geq 0$ ,*

$$\int_0^1 B_n(x) F(x) dx = \frac{B_{n+1}(1) - \mathbb{E}B_{n+1}(Y)}{n+1}. \quad (70)$$

The expectation is an explicit rational combination of the moments in equation (69). Its exponential generating function is

$$\sum_{n=0}^{\infty} \mathbb{E}B_n(Y) \frac{t^n}{n!} = \frac{t}{e^t - 1} \mathcal{A}(t). \quad (71)$$

Symmetry implies  $\mathbb{E}B_n(Y) = 0$  for every odd  $n$ .

*Proof.* Use  $B'_{n+1} = (n+1)B_n$  and integrate by parts against  $F$ . The generating function is the classical generating function of Bernoulli polynomials averaged at  $Y$ . Finally,  $B_n(1-x) = (-1)^n B_n(x)$  and  $Y \stackrel{d}{=} 1 - Y$ .  $\square$

## 6.4 Moments of ordinary derivatives

**Theorem 6.6** (Derivative moments). *For  $n \geq 1$ , the Mellin transform of  $F^{(n)}$  is the entire identity*

$$\int_0^1 x^p F^{(n)}(x) dx = (-1)^{n-1} p^{n-1} \mu(p-n+1), \quad p \in \mathbb{C}. \quad (72)$$

For integer  $p < n-1$ , the falling factorial vanishes. For integers  $p, n \geq 0$ ,

$$\int_{-1}^1 x^p \text{up}^{(n)}(x) dx = \begin{cases} (-1)^n p^n m_{p-n}, & p \geq n, \\ 0, & p < n. \end{cases} \quad (73)$$

These are [proved here](#).

*Proof.* Integrate by parts until only  $F'$  remains; every higher derivative vanishes at both endpoints. The last integral is  $\mu(p-n+1)$ . Flatness makes both sides entire in  $p$ . The compactly supported up formula follows from  $n$  integrations by parts.  $\square$

## 6.5 Beta integrals and products

**Proposition 6.7** (Probability-integral-transform beta family). *For every integrable  $\varphi$ ,*

$$\int_0^1 \varphi(F(x)) F'(x) dx = \int_0^1 \varphi(u) du. \quad (74)$$

Therefore, for  $\text{Re } a, \text{Re } b > -1$ ,

$$\int_0^1 F(x)^a (1-F(x))^b F'(x) dx = B(a+1, b+1). \quad (75)$$

On the left half,

$$\int_0^{1/2} F(x)^a \text{up}(x)^b F(2x) dx = \frac{1}{2} B_{1/2}(a+1, b+1). \quad (76)$$

If  $H(z) = F((z+1)/2)$  is the CDF of up, then

$$\int_{-1}^1 \text{up}(z) H(z)^a (1-H(z))^b dz = B(a+1, b+1). \quad (77)$$

*Proof.* Use  $u = F(x)$  in equation (74); then use  $F' = 2F(2x)$  on the left half. The last formula is the same probability-integral transform for  $H$ .  $\square$

**Proposition 6.8** (Powers of the CDF and order statistics). *For  $m > 0$  and  $0 \leq x \leq 1$ ,*

$$\int_0^x F(t)^m dt = xF(x)^m - m \int_0^{F(x)} u^{m-1} G(u) du. \quad (78)$$

*If  $m \in \mathbb{N}$  and  $Y_1, \dots, Y_m$  are independent copies of  $Y$ , then*

$$\int_0^1 F(x)^m dx = 1 - \mathbb{E} \max(Y_1, \dots, Y_m) = \int_0^1 (1 - F(x))^m dx. \quad (79)$$

*Proof.* Integrate  $F^m$  by parts and substitute  $u = F(t)$ . The CDF of the maximum is  $F^m$ ; symmetry gives the last equality.  $\square$

## 6.6 A mixed product and a sinc-product integral

**Theorem 6.9** (Gini–Fourier identity). *Let*

$$\Phi(t) = \widehat{\text{up}}(t) = \prod_{j=1}^{\infty} \text{sinc}(t/2^j). \quad (80)$$

*Then*

$$\int_0^1 F(x)(1 - F(x)) dx = \frac{1}{2} \mathbb{E}|Y_1 - Y_2| \quad (81)$$

$$= \frac{1}{4} \mathbb{E}|Z_1 - Z_2| \quad (82)$$

$$= \frac{1}{2\pi} \int_0^{\infty} \frac{1 - \Phi(t)^2}{t^2} dt. \quad (83)$$

*Numerically,*

$$\int_0^1 F(x)(1 - F(x)) dx = 0.095563232016\dots \quad (84)$$

*The identities are **proved here**; the decimal is **numerically checked**.*

*Proof.* For every continuous CDF  $C$ ,  $\mathbb{E}|X_1 - X_2| = 2 \int C(1 - C)$ . The affine relation  $Z = 2Y - 1$  gives the second line. Finally,  $\mathbb{E}|W| = (2/\pi) \int_0^{\infty} (1 - \text{Re } \varphi_W(t)) t^{-2} dt$ , while the characteristic function of  $Z_1 - Z_2$  is  $\Phi(t)^2$ .  $\square$

## 7 Laplace, Fourier, Mellin, Cauchy, and correlation transforms

### 7.1 Laplace transforms of $F$ and up

Define the negative moment transform

$$\phi(s) = \mathbb{E} e^{-sY} = \prod_{j=1}^{\infty} \frac{1 - e^{-s/2^j}}{s/2^j}, \quad \text{Re } s \geq 0. \quad (85)$$

**Theorem 7.1** (Laplace dictionary). *For  $\operatorname{Re} s > 0$ ,*

$$\int_0^\infty e^{-sx} F(x) dx = \frac{\phi(s)}{s}, \quad (86)$$

$$\int_0^1 e^{-sx} F(x) dx = \frac{\phi(s) - e^{-s}}{s}, \quad (87)$$

$$\int_0^1 e^{-sx} \operatorname{up}(x) dx = \frac{1 - \phi(s)}{s}. \quad (88)$$

*The bilateral moment and Fourier transforms of  $\operatorname{up}$  are*

$$\int_{-1}^1 e^{sx} \operatorname{up}(x) dx = \prod_{j=1}^{\infty} \frac{\sinh(s/2^j)}{s/2^j}, \quad (89)$$

$$\widehat{\operatorname{up}}(t) = \prod_{j=1}^{\infty} \operatorname{sinc}(t/2^j). \quad (90)$$

*Proof.* Integration by parts against the probability measure  $dF$  gives  $\mathcal{L}F = \phi/s$ . Subtract the tail  $\int_1^\infty e^{-sx} dx$  for the unit interval and use  $\operatorname{up} = 1 - F$  on  $[0, 1]$ . Independence in equation (8) gives the products for  $Z$ .  $\square$

**Corollary 7.2** (Transforms of fractional primitives and derivatives). *For  $\operatorname{Re} \alpha > 0$ ,*

$$\mathcal{L}\{I_{0+}^\alpha F\}(s) = s^{-\alpha-1} \phi(s). \quad (91)$$

*For noninteger  $\alpha > 0$ , because all initial data of  $F$  vanish at 0,*

$$\mathcal{L}\{D_{0+}^\alpha F\}(s) = s^{\alpha-1} \phi(s). \quad (92)$$

*Thus Bromwich inversion gives*

$$I_{0+}^\alpha F(x) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} s^{-\alpha-1} \phi(s) e^{sx} ds, \quad (93)$$

$$D_{0+}^\alpha F(x) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} s^{\alpha-1} \phi(s) e^{sx} ds. \quad (94)$$

## 7.2 Laplace and Mellin transforms of the inverse

**Theorem 7.3** (Inverse-transform dictionary). *For  $\operatorname{Re} s > 0$ ,*

$$\int_0^1 e^{-sy} G(y) dy = \frac{\int_0^1 e^{-sF(x)} dx - e^{-s}}{s}, \quad (95)$$

$$\int_0^1 e^{-sy} G(y)^q dy = \frac{q}{s} \int_0^1 x^{q-1} (e^{-sF(x)} - e^{-s}) dx, \quad \operatorname{Re} q > 0. \quad (96)$$

*For  $\operatorname{Re} s > 0$ ,*

$$\int_0^1 y^{s-1} G(y) dy = \frac{1 - \int_0^1 F(x)^s dx}{s}. \quad (97)$$

*All formulas are [proved here](#).*

*Proof.* Integrate by parts, then use  $G'(y) dy = dx$  under  $y = F(x)$ . For the power formula, write the boundary term  $e^{-s}$  as  $qe^{-s} \int_0^1 x^{q-1} dx$ .  $\square$

### 7.3 Stieltjes and Cauchy transforms

**Proposition 7.4** (Stieltjes transform of the Fabius law). *For  $z \in \mathbb{C} \setminus [0, 1]$ ,*

$$S_Y(z) = \int_0^1 \frac{dF(x)}{z - x} = \int_0^1 \frac{dy}{z - G(y)} = \sum_{n=0}^{\infty} \frac{\mu_n}{z^{n+1}}, \quad |z| > 1. \quad (98)$$

*If  $C_{\text{up}}(w) = \int_{-1}^1 \text{up}(t)/(w - t) dt$ , then*

$$S_Y(z) = 2C_{\text{up}}(2z - 1). \quad (99)$$

**Proposition 7.5** (Ordinary Cauchy transforms). *For  $z \in \mathbb{C} \setminus [0, 1]$ , with a consistent logarithm branch,*

$$\int_0^1 \frac{F(x)}{z - x} dx = \mathbb{E} \log \left( \frac{z - Y}{z - 1} \right) = \sum_{n=1}^{\infty} \frac{1 - \mu_n}{nz^n}, \quad |z| > 1, \quad (100)$$

$$\int_0^1 \frac{G(y)}{z - y} dy = \int_0^1 \log \left( \frac{z - F(x)}{z - 1} \right) dx. \quad (101)$$

*Proof.* For equation (100), integrate by parts with antiderivative  $-\log(z - x)$ , then expand the logarithms at infinity. The inverse formula is identical, followed by  $y = F(x)$ .  $\square$

### 7.4 Autocorrelation, convolution, and Sobolev energies

**Proposition 7.6** (Autocorrelation). *The autocorrelation*

$$C(a) = \int_{\mathbb{R}} \text{up}(x) \text{up}(x + a) dx \quad (102)$$

*has the Fourier representation*

$$C(a) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{ita} \Phi(t)^2 dt. \quad (103)$$

*More generally, every  $m$ -fold convolution of  $\text{up}$  has transform  $\Phi^m$  and is the density of a sum of  $m$  independent copies of  $Z$ .*

**Definition 7.7** (Fractional Sobolev energy). For  $\alpha \geq 0$ , define

$$\mathcal{E}_{\alpha} = \int_{\mathbb{R}} \text{up}(x) (-\Delta)^{\alpha/2} \text{up}(x) dx = \frac{1}{\pi} \int_0^{\infty} t^{\alpha} \Phi(t)^2 dt. \quad (104)$$

The rapid decay of  $\Phi$  makes this finite for every  $\alpha \geq 0$ .

**Proposition 7.8** (Energy refinement identity and log-convexity). *The product relation  $\Phi(2t) = \text{sinc}(t)\Phi(t)$  implies*

$$\mathcal{E}_{\alpha} = \frac{2^{\alpha+1}}{\pi} \int_0^{\infty} t^{\alpha} \text{sinc}(t)^2 \Phi(t)^2 dt. \quad (105)$$

*The map  $\alpha \mapsto \mathcal{E}_{\alpha}$  is log-convex on  $[0, \infty)$ .*

*Proof.* Substitute  $u = 2t$  in equation (104). Log-convexity is Holder's inequality applied to the positive measure  $\Phi(t)^2 dt/\pi$  and the functions  $t^{\alpha}$ .  $\square$

The accompanying computation gives

|                        |                |                |               |               |                |
|------------------------|----------------|----------------|---------------|---------------|----------------|
| $\alpha$               | 0              | 1/2            | 1             | 2             | 4              |
| $\mathcal{E}_{\alpha}$ | 0.808873535968 | 0.939432032833 | 1.28502241154 | 3.23549414387 | 51.7679063017. |

(106)

These decimals are **numerically checked**.

## 8 Riemann–Liouville and Caputo calculus on dyadic scales

### 8.1 The exact order-shift equation

**Theorem 8.1** (Fractional order shift). *For  $\operatorname{Re} \alpha > 0$  and  $0 \leq x \leq 1$ ,*

$$\boxed{\mathcal{J}_{\alpha+1}(x) = 2^\alpha \mathcal{J}_\alpha(x/2)}. \quad (107)$$

*For  $\mathcal{F}$ , the same identity holds for every real  $x$ . This is [proved here](#).*

*Proof.* Since  $F(t) = \frac{d}{dt}F(t/2)$  for  $0 \leq t \leq 1$ , integration by parts gives

$$\mathcal{J}_{\alpha+1}(x) = \frac{1}{\Gamma(\alpha)} \int_0^x (x-t)^{\alpha-1} F(t/2) dt.$$

Set  $t = 2u$ . The global version uses  $\mathcal{F}(t) = \frac{d}{dt}\mathcal{F}(t/2)$  for all  $t$ . □

**Corollary 8.2** (Integer shift of a fractional base order). *For  $n \in \mathbb{N}_0$ ,  $\operatorname{Re} \beta > 0$ , and  $0 \leq x \leq 1$ ,*

$$\mathcal{J}_{n+\beta}(x) = 2^{n\beta + \binom{n}{2}} \mathcal{J}_\beta(x/2^n). \quad (108)$$

### 8.2 Endpoint values and the fractional dyadic lattice

**Theorem 8.3** (Fractional endpoint moment). *For  $\operatorname{Re} \alpha > 0$ ,*

$$\mathcal{J}_\alpha(1) = \frac{\mu(\alpha)}{\Gamma(\alpha+1)}. \quad (109)$$

*Consequently, for  $n \in \mathbb{N}_0$  and  $\operatorname{Re} \beta > 0$ ,*

$$\boxed{\mathcal{J}_\beta(2^{-n}) = 2^{-n\beta - \binom{n}{2}} \frac{\mu(n+\beta)}{\Gamma(n+\beta+1)}}. \quad (110)$$

*These identities are [proved here](#).*

*Proof.* Symmetry transforms the defining integral into

$$\mathcal{J}_\alpha(1) = \frac{1}{\Gamma(\alpha)} \int_0^1 u^{\alpha-1} (1-F(u)) du.$$

The tail-moment formula  $\mu(\alpha) = \alpha \int_0^1 u^{\alpha-1} (1-F(u)) du$  proves the first claim. Put  $x = 1$  in equation (108) and solve for  $\mathcal{J}_\beta(2^{-n})$ . □

*Remark 8.4.* At  $\beta = 0$ , interpreted by continuity, equation (110) becomes the classical integer-moment dyadic formula equation (23). The new point is that the lattice extends continuously and analytically in the order.

### 8.3 Fractional calculus for up

**Theorem 8.5** (Up–Fabius fractional bridge). *For  $\operatorname{Re} \alpha > 0$  and  $x \geq -1$ ,*

$$\mathcal{K}_{\alpha+1}(x) = 2^\alpha \mathcal{J}_\alpha\left(\frac{x+1}{2}\right). \quad (111)$$

*At the right support endpoint,*

$$\mathcal{K}_\alpha(1) = 2^{\alpha-1} \frac{\mu(\alpha-1)}{\Gamma(\alpha)}, \quad \alpha > 0, \quad (112)$$

*where the moment is understood by its entire continuation. These are **proved here**.*

*Proof.* The function  $H(x) = F((x+1)/2)$  is the CDF of up, hence  $H' = \text{up}$ . Integrate the  $\mathcal{K}_{\alpha+1}$  kernel by parts and substitute  $u = (t+1)/2$ . At  $x = 1$ , alternatively write

$$\mathcal{K}_\alpha(1) = \Gamma(\alpha)^{-1} \mathbb{E}(1-Z)^{\alpha-1} = 2^{\alpha-1} \Gamma(\alpha)^{-1} \mathbb{E}(1-Y)^{\alpha-1}$$

and use symmetry. □

**Corollary 8.6** (Shifted fractional dyadic lattice). *For  $n \in \mathbb{N}_0$  and  $\operatorname{Re} \beta > 0$ ,*

$$\mathcal{K}_{\beta+1}(-1 + 2^{1-n}) = 2^{\beta-n\beta-\binom{n}{2}} \frac{\mu(n+\beta)}{\Gamma(n+\beta+1)}. \quad (113)$$

### 8.4 Fractional derivatives of $F$

**Theorem 8.7** (Fractional derivative on the first dyadic cell). *For  $0 < \beta < 1$  and  $0 \leq x \leq 1/2$ ,*

$$D_{0+}^\beta F(x) = {}^C D_{0+}^\beta F(x) = 2^\beta \mathcal{J}_{1-\beta}(2x). \quad (114)$$

*At dyadic points,*

$$D_{0+}^\beta F(2^{-n-1}) = 2^{\beta-n(1-\beta)-\binom{n}{2}} \frac{\mu(n+1-\beta)}{\Gamma(n+2-\beta)}. \quad (115)$$

*The formulas are **proved here**.*

*Proof.* Endpoint flatness gives  $D^\beta F = I^{1-\beta} F'$ . On  $[0, 1/2]$ , insert  $F'(t) = 2F(2t)$  and put  $u = 2t$ . Then use equation (110). □

**Corollary 8.8** (General mixed integer/fractional derivative). *Let  $q = m + \beta$  with  $m \in \mathbb{N}_0$  and  $0 < \beta < 1$ . For  $0 \leq x \leq 2^{-m-1}$ ,*

$$D_{0+}^{m+\beta} F(x) = 2^{\binom{m+1}{2} + m\beta} D_{0+}^\beta F(2^m x). \quad (116)$$

*Combining with equation (115) gives an explicit moment formula on every compatible dyadic lattice.*

## 8.5 Complements and right-sided fractional integrals

**Proposition 8.9** (Fractional complements). *For  $0 \leq x \leq 1$  and  $\operatorname{Re} \alpha > 0$ ,*

$$I_{0+}^{\alpha}[1 - F](x) = \frac{x^{\alpha}}{\Gamma(\alpha + 1)} - \mathcal{J}_{\alpha}(x), \quad (117)$$

$$I_{1-}^{\alpha}F(x) = \frac{(1 - x)^{\alpha}}{\Gamma(\alpha + 1)} - \mathcal{J}_{\alpha}(1 - x), \quad (118)$$

$$I_{1-}^{\alpha}[1 - F](x) = \mathcal{J}_{\alpha}(1 - x). \quad (119)$$

Here  $I_{1-}^{\alpha}f(x) = \Gamma(\alpha)^{-1} \int_x^1 (t - x)^{\alpha-1} f(t) dt$ .

*Proof.* The first formula is linearity. In the right-sided integrals put  $u = 1 - t$  and use  $F(1 - u) = 1 - F(u)$ .  $\square$

## 9 A continuous dyadic Appell family and its exact fractional defect

This section contains the central frontier result. Integer integration and differentiation fit a natural continuous dilation interpolation. The interpolation is almost, but not exactly, Riemann–Liouville calculus. The discrepancy is the repository’s Gamma–zeta periodic term.

### 9.1 The continuous-order dyadic family

**Definition 9.1** (Dyadic Appell family). For  $q \in \mathbb{R}$  and any function  $f$  on  $\mathbb{R}$ , define

$$(\mathfrak{A}_q f)(x) = 2^{q(q-1)/2} f(2^{-q}x). \quad (120)$$

We apply this operator principally to  $f = \mathcal{F}$ .

**Theorem 9.2** (Interpolation, derivative ladder, and cocycle). *For real  $q, \alpha, \beta$ :*

$$\mathfrak{A}_n \mathcal{F} = I_{0+}^n \mathcal{F} \quad (n \in \mathbb{N}_0), \quad (121)$$

$$\mathfrak{A}_{-n} \mathcal{F} = \mathcal{F}^{(n)} \quad (n \in \mathbb{N}_0), \quad (122)$$

$$\frac{d}{dx} \mathfrak{A}_q \mathcal{F} = \mathfrak{A}_{q-1} \mathcal{F}, \quad (123)$$

$$\mathfrak{A}_{\alpha} \mathfrak{A}_{\beta} = 2^{-\alpha\beta} \mathfrak{A}_{\alpha+\beta}. \quad (124)$$

All statements are *proved here*.

*Proof.* The integer identities are equations (21) and (24). Differentiate equation (120) and use  $\mathcal{F}' = 2\mathcal{F}(2\cdot)$ . Direct composition of the two dilations gives the scalar cocycle  $2^{-\alpha\beta}$ .  $\square$

*Remark 9.3.* The cocycle is not the Riemann–Liouville semigroup law. Thus  $q \mapsto \mathfrak{A}_q$  is an Appell interpolation of a single function’s integer ladder, not a fractional-integration semigroup on an arbitrary function space. The next theorem quantifies the difference exactly.

## 9.2 Laplace transform of the signed continuation

Let

$$\mathcal{L}(s) = \int_0^\infty e^{-sx} \mathcal{F}(x) dx, \quad s > 0. \quad (125)$$

**Theorem 9.4** (Thue–Morse Laplace completion). *For  $s > 0$ ,*

$$\mathcal{L}(s) = e^{-s} \left( \prod_{j=1}^\infty \frac{\sinh(s/2^j)}{s/2^j} \right) \left( \prod_{j=0}^\infty (1 - e^{-2^{j+1}s}) \right) \quad (126)$$

$$= \phi(2s) \prod_{j=0}^\infty (1 - e^{-2^{j+1}s}). \quad (127)$$

*It satisfies the exact functional equation*

$$\mathcal{L}(s/2) = s\mathcal{L}(s). \quad (128)$$

*The statements are [proved here](#).*

*Proof.* On the block  $[2n, 2n+2]$ ,  $\mathcal{F}(x) = \varepsilon_n \operatorname{up}(x - 2n - 1)$ . Summing the block Laplace transforms gives

$$\mathcal{L}(s) = e^{-s} M_Z(-s) \sum_{n \geq 0} \varepsilon_n e^{-2ns}.$$

Evenness makes  $M_Z(-s) = M_Z(s)$ , and the Thue–Morse generating product is  $\sum \varepsilon_n q^n = \prod_{j \geq 0} (1 - q^{2^j})$ . Since  $e^{-s} M_Z(s) = \phi(2s)$ , both products follow. Alternatively, Laplace-transforming  $\mathcal{F}' = 2\mathcal{F}(2x)$  gives  $s\mathcal{L}(s) = \mathcal{L}(s/2)$  because  $\mathcal{F}(0) = 0$ .  $\square$

## 9.3 Exact completion of the periodic endpoint term

Put  $L = \log 2$ . The audited repository proves, for the same  $\phi$ , an exact Lambert– $W$ /Gamma–zeta endpoint decomposition [12, 13, 11]:

$$\log \phi(s) = -\frac{(\log s)^2}{2L} + \frac{1}{2} \log s + P(\log_2 s) + E(s), \quad (129)$$

where  $P$  is smooth and one-periodic and

$$E(s) = -\sum_{j=0}^\infty \log(1 - e^{-2^j s}). \quad (130)$$

The repository also proves that the centered function  $\Psi = P - \int_0^1 P$  has Fourier coefficients

$$\widehat{\Psi}(k) = -\frac{\Gamma(-\chi_k) \zeta(1 - \chi_k)}{L}, \quad \chi_k = \frac{2\pi i k}{L}, \quad k \neq 0, \quad (131)$$

and every one of them is nonzero.

**Theorem 9.5** (Exact bilateral completion). *For every  $s > 0$ ,*

$$\boxed{\log \mathcal{L}(s) = -\frac{(\log s)^2}{2L} - \frac{1}{2} \log s + P(\log_2 s).} \quad (132)$$

*This is [derived from audited inputs](#) (using the audited decomposition).*

*Proof.* Apply equation (129) to  $2s$ . Periodicity gives  $P(\log_2(2s)) = P(\log_2 s)$ , and the quadratic terms simplify to  $-(\log s)^2/(2L) - (\log s)/2$ . By equation (130),  $E(2s)$  is the negative logarithm of the second product in equation (127), so it cancels exactly.  $\square$

## 9.4 The exact Riemann–Liouville defect

The ideal order- $q$  Riemann–Liouville action on a flat function has Laplace multiplier  $s^{-q}$ . The actual transform of the dyadic interpolation is

$$\mathcal{L}\{\mathfrak{A}_q \mathcal{F}\}(s) = 2^{q(q+1)/2} \mathcal{L}(2^q s). \quad (133)$$

**Theorem 9.6** (Gamma–zeta fractional-dilation defect). *For  $q \in \mathbb{R}$  and  $s > 0$ , define*

$$R_q(s) = \frac{\mathcal{L}\{\mathfrak{A}_q \mathcal{F}\}(s)}{s^{-q} \mathcal{L}(s)}. \quad (134)$$

*Then*

$$R_q(s) = \exp(P(\log_2 s + q) - P(\log_2 s)) = \exp(\Psi(\log_2 s + q) - \Psi(\log_2 s)). \quad (135)$$

*Moreover,*

$$R_{q+1}(s) = R_q(s), \quad (136)$$

$$R_n(s) = 1 \quad (n \in \mathbb{Z}), \quad (137)$$

$$R_q(s) \equiv 1 \text{ for all } s > 0 \iff q \in \mathbb{Z}. \quad (138)$$

*The theorem is **derived from audited inputs** (using theorem 9.5 and the audited nonvanishing of the Gamma–zeta Fourier coefficients).*

*Proof.* Use equations (132) and (133); the quadratic and linear logarithmic terms cancel exactly, leaving the difference of  $P$ . Periodicity of  $P$  gives equations (136) and (137). If  $R_q \equiv 1$ , then  $\Psi(u + q) = \Psi(u)$  for every  $u$ . The Fourier mode  $k = 1$  is nonzero by equation (131), so  $e^{2\pi i q} = 1$  and  $q \in \mathbb{Z}$ . The converse was already shown.  $\square$

**Corollary 9.7** (Uniform defect bounds). *Let  $\text{osc } P = \max P - \min P$ . Then*

$$e^{-\text{osc } P} \leq R_q(s) \leq e^{\text{osc } P} \quad (q \in \mathbb{R}, s > 0). \quad (139)$$

*The numerical Gamma–zeta series gives*

$$\text{osc } P = 4.2507727262 \dots \times 10^{-6}, \quad (140)$$

*so the natural dyadic interpolation is extraordinarily close to fractional calculus, but rigorously never identical at a noninteger order.*

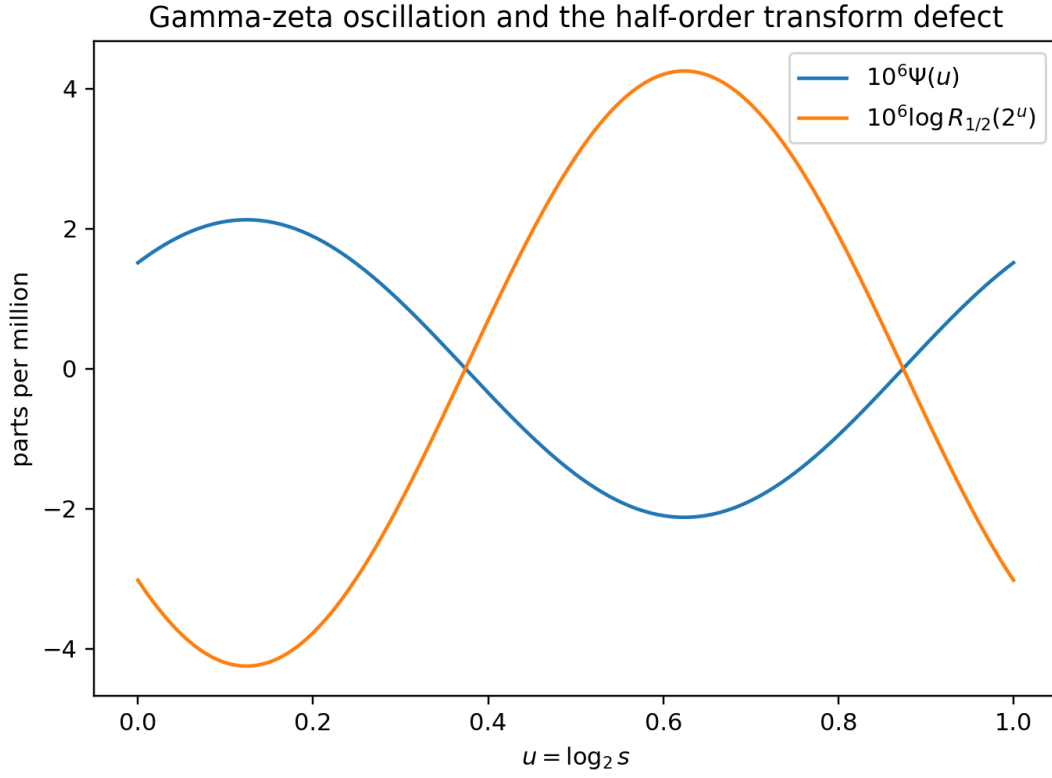


Figure 1: The centered Gamma–zeta oscillation  $\Psi(u)$  and the exact half-order defect  $\log R_{1/2}(2^u) = \Psi(u + 1/2) - \Psi(u)$ . The vertical scale is in parts per million.

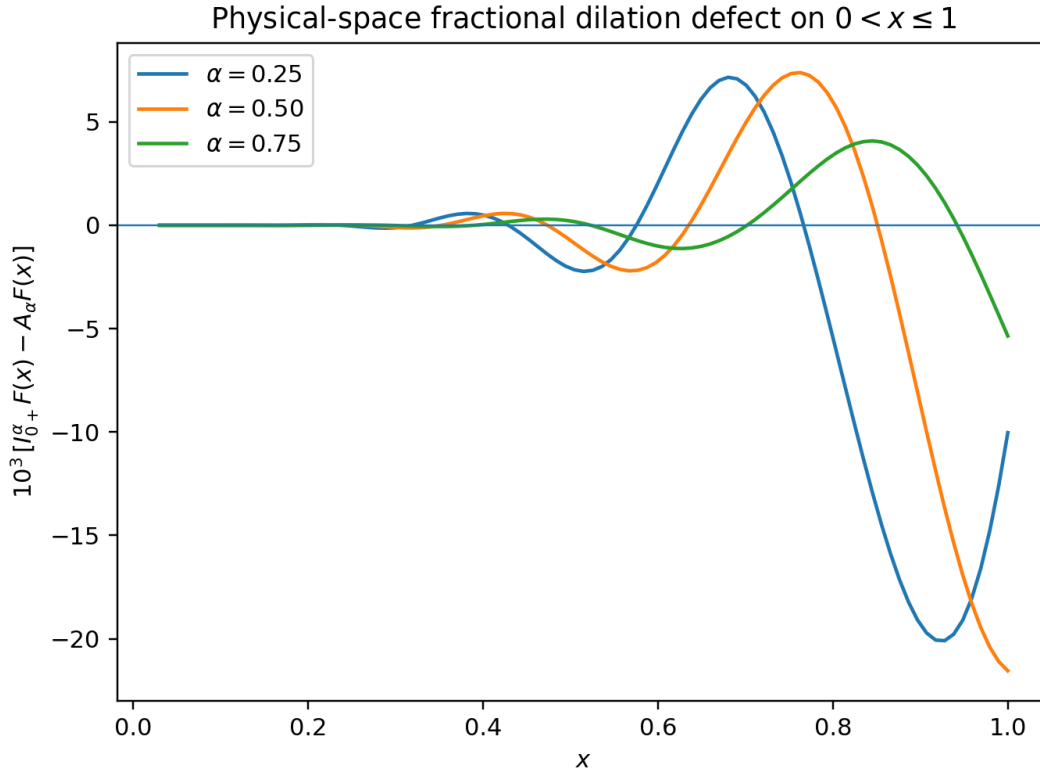


Figure 2: Physical-space comparison of a Riemann–Liouville fractional integral of  $F$  with the continuous dyadic interpolation. The visible difference is rescaled; the underlying identity is exact only at integer order.

## 9.5 Interpretation

The defect theorem gives a precise answer to a natural but previously ambiguous question: what should “half an antiderivative” mean for the Fabius dilation equation? There are two canonical answers.

- Riemann–Liouville calculus preserves the additive order semigroup and has multiplier  $s^{-q}$ .
- Dyadic Appell interpolation preserves the exact Fabius dilation shape and the derivative ladder  $\partial_x \mathfrak{A}_q = \mathfrak{A}_{q-1}$ .

They agree on the integer lattice. Away from it, their ratio is not an uncontrolled error but a smooth log-periodic multiplier with explicit Gamma–zeta Fourier coefficients. The same periodic object already controls the sharp Lambert- $W$  endpoint asymptotics; it therefore acts as an obstruction simultaneously in physical endpoint asymptotics and in continuous-order operator interpolation.

## 10 Whole-line fractional operators acting on up

The left-sided operators remember a base point. Translation-invariant fractional calculus is better expressed through the Riesz fractional Laplacian and potential.

## 10.1 Fractional Laplacian outside the support

For  $0 < \alpha < 2$ , use the standard singular-integral normalization [9]:

$$(-\Delta)^{\alpha/2} f(x) = C_{1,\alpha} \text{PV} \int_{\mathbb{R}} \frac{f(x) - f(y)}{|x - y|^{1+\alpha}} dy, \quad C_{1,\alpha} = \frac{2^\alpha \Gamma((1+\alpha)/2)}{\sqrt{\pi} |\Gamma(-\alpha/2)|}. \quad (141)$$

**Theorem 10.1** (Exact exterior moment series). *For  $0 < \alpha < 2$  and  $|x| > 1$ ,*

$$(-\Delta)^{\alpha/2} \text{up}(x) = -C_{1,\alpha} \sum_{j=0}^{\infty} \frac{(1+\alpha)_{2j}}{(2j)!} \frac{m_{2j}}{|x|^{1+\alpha+2j}}. \quad (142)$$

*The series converges absolutely and locally uniformly on  $|x| > 1$ . This is [proved here](#).*

*Proof.* Outside the support,  $\text{up}(x) = 0$ , so

$$(-\Delta)^{\alpha/2} \text{up}(x) = -C_{1,\alpha} \int_{-1}^1 \frac{\text{up}(y)}{|x - y|^{1+\alpha}} dy.$$

For  $x > 1$ , expand  $(x - y)^{-1-\alpha} = x^{-1-\alpha} \sum_{k \geq 0} (1+\alpha)_k (y/x)^k / k!$ . Odd moments vanish. The case  $x < -1$  is identical by evenness.  $\square$

The first terms are

$$(-\Delta)^{\alpha/2} \text{up}(x) = -C_{1,\alpha} |x|^{-1-\alpha} \left( 1 + \frac{(1+\alpha)(2+\alpha)}{18x^2} + \frac{19(1+\alpha)_4}{16200x^4} + \dots \right). \quad (143)$$

At  $x = 2$ , six even moments reproduce direct quadrature to roughly  $10^{-7}$  for the tested orders  $\alpha = 0.5, 0.7, 1.3$ ; this is [numerically checked](#).

## 10.2 Riesz potentials

For  $0 < \alpha < 1$ , define the one-dimensional Riesz potential with Fourier multiplier  $|t|^{-\alpha}$  by

$$I_\alpha^R f(x) = c_{1,\alpha} \int_{\mathbb{R}} |x - y|^{\alpha-1} f(y) dy, \quad c_{1,\alpha} = \frac{\Gamma((1-\alpha)/2)}{2^\alpha \sqrt{\pi} \Gamma(\alpha/2)}. \quad (144)$$

**Theorem 10.2** (Exterior Riesz-potential series). *For  $0 < \alpha < 1$  and  $|x| > 1$ ,*

$$I_\alpha^R \text{up}(x) = c_{1,\alpha} |x|^{\alpha-1} \sum_{j=0}^{\infty} \frac{(1-\alpha)_{2j}}{(2j)!} \frac{m_{2j}}{|x|^{2j}}. \quad (145)$$

*The series is absolutely and locally uniformly convergent.*

*Proof.* Expand  $|x - y|^{\alpha-1} = |x|^{\alpha-1} (1 - \text{sgn}(x)y/|x|)^{-(1-\alpha)}$  and use evenness.  $\square$

## 10.3 Transform-side fractional derivatives

By Fourier multiplication,

$$\widehat{(-\Delta)^{\alpha/2} \text{up}(t)} = |t|^\alpha \prod_{j=1}^{\infty} \text{sinc}(t/2^j). \quad (146)$$

This supplies a direct integral transform for all translation-invariant fractional derivatives of  $\text{up}$ . It also makes the energies in equation (104) the diagonal quadratic form of the operator.

## 11 The inverse Mellin transform has a Gevrey-2 boundary

This section uses only the audited endpoint quadratic law

$$-\log F(e^{-u}) \sim \frac{u^2}{2 \log 2}, \quad u \rightarrow \infty. \quad (147)$$

It yields a new regularity classification for the inverse Mellin transform.

Define

$$A(s) = \int_0^1 F(x)^s dx, \quad \operatorname{Re} s > 0, \quad (148)$$

and

$$Q(x) = -\log F(x), \quad a_n = \int_0^1 Q(x)^n dx. \quad (149)$$

By equation (97),

$$\mathcal{M}_G(s) := \int_0^1 y^{s-1} G(y) dy = \frac{1 - A(s)}{s}. \quad (150)$$

**Theorem 11.1** (Sharp logarithmic-moment growth). *The moments of  $Q = -\log F$  satisfy*

$$\lim_{n \rightarrow \infty} \left( \frac{a_n}{(2n)!} \right)^{1/n} = \frac{1}{2 \log 2}. \quad (151)$$

*This is **derived from audited inputs** (using the audited endpoint quadratic law).*

*Proof.* Put  $x = e^{-u}$ . Then

$$a_n = \int_0^\infty Q(e^{-u})^n e^{-u} du.$$

Let  $c = (2 \log 2)^{-1}$ . For every  $\varepsilon > 0$ , there is  $U$  such that  $(c - \varepsilon)u^2 \leq Q(e^{-u}) \leq (c + \varepsilon)u^2$  for  $u \geq U$ . The upper bound splits the integral at  $U$  and compares the tail with  $(c + \varepsilon)^n \Gamma(2n + 1)$ . The compact part is exponentially negligible after division by  $(2n)!$ . The lower bound compares with the incomplete gamma tail  $(c - \varepsilon)^n \int_U^\infty u^{2n} e^{-u} du$ , whose ratio to  $(2n)!$  tends to 1. Take  $n$ th roots and then  $\varepsilon \downarrow 0$ .  $\square$

**Theorem 11.2** (Gevrey-2 but nonanalytic inverse Mellin boundary). *The functions  $A(s)$  and  $\mathcal{M}_G(s)$  have derivatives of every order from the right at  $s = 0$ , with*

$$A^{(n)}(0+) = (-1)^n a_n, \quad (152)$$

$$\mathcal{M}_G^{(n)}(0+) = -\frac{A^{(n+1)}(0+)}{n+1}. \quad (153)$$

*Their formal Taylor series at 0 have radius of convergence zero. More precisely, the derivative growth is of Gevrey order 2 and of no smaller Gevrey order in the root-growth sense. Consequently neither transform has a holomorphic continuation through  $s = 0$  that agrees with its defining integral on  $\operatorname{Re} s > 0$ . This is **derived from audited inputs**.*

*Proof.* Every  $a_n$  is finite by endpoint flatness, so differentiation under the integral from the right gives equation (152). Divide the Taylor identity  $1 - A(s)$  by  $s$  to obtain equation (153). By equation (151),  $a_n$  is asymptotic in root scale to  $c^n (2n)!$ , and  $(2n)!$  is of order  $4^n (n!)^2$  up to subexponential factors. Hence the derivatives have Gevrey-2 growth. The Taylor coefficients  $a_n/n!$  have unbounded  $n$ th root, so the radius is zero. A holomorphic continuation would have a positive Taylor radius, a contradiction.  $\square$

*Remark 11.3.* This theorem explains a sharp asymmetry. The direct moment  $\mu(s) = \mathbb{E}Y^s$  is entire because it integrates against the flat density  $F'$ . The inverse Mellin transform probes powers of  $F$  itself; at  $s = 0$  it sees the full quadratic logarithmic endpoint profile and becomes only Gevrey-2. The Lambert- $W$  endpoint geometry is therefore visible not only pointwise in  $G$  but also as a transform-domain analyticity barrier.

## 12 Numerical experiments and reproducibility

The accompanying program `numerical_experiments.py` is a verification layer, not a source of proof. It uses three independent representations:

1. exact rational cumulants and moments generated from Bernoulli numbers;
2. a finite weighted Irwin–Hall spline for  $\sum_{j=1}^N 2^{-j}V_j$ , with exact Thue–Morse inclusion–exclusion signs and the unresolved tail replaced by its mean;
3. infinite Laplace/Fourier products and the Gamma–zeta Fourier series of  $P$ .

At spline level  $N = 11$ , the tests give:

- ordinary primitive errors below  $2.0 \times 10^{-8}$  at the sampled points;
- fractional shift errors below  $1.3 \times 10^{-8}$ ;
- fractional dyadic-lattice errors below  $1.3 \times 10^{-8}$ ;
- direct versus transmuted inverse fractional integrals agreeing to about  $2.2 \times 10^{-10}$ ;
- the mixed product value in equation (84);
- exterior fractional-Laplacian moment series agreeing with direct quadrature to  $10^{-7}$ – $10^{-6}$  using six terms;
- the exact integer-order defect  $R_n = 1$  and the order periodicity  $R_{q+1} = R_q$  to the working precision.

Representative half-order values are

$$\begin{aligned} R_{1/2}(0.1) &= 1.000004008430907\dots, \\ R_{1/2}(1) &= 0.999996976545554\dots, \\ R_{1/2}(10) &= 0.999998632323868\dots \end{aligned} \tag{154}$$

The complete machine-readable output is included as `numerical_results.txt`.

### 12.1 Numerical cautions

The finite-spline formula is an alternating sum with severe cancellation if evaluated too far on the right. The code therefore computes only on the stable left half and imposes the exact reflection  $F(1-x) = 1 - F(x)$ . Fractional kernels are desingularized by substitutions before adaptive quadrature. The Gamma–zeta defect is evaluated at high precision because its amplitude is only a few parts in  $10^6$ .

No decimal in this report is used to assert equality. Each exact identity has an independent symbolic proof in the preceding sections.

## 13 Conjectures and research directions

### 13.1 Higher inverse derivatives

The first inverse derivative has the sharp threshold in theorem 5.8. The Lambert endpoint heuristic suggests a complete hierarchy.

**Conjecture 13.1** (Higher inverse-derivative threshold). *For every integer  $n \geq 1$  and  $r > 0$ ,*

$$G^{(n)} \in L^r(0, 1) \iff 0 < r \leq \frac{1}{n}. \quad (155)$$

*conjectural/open*

A formal endpoint differentiation of  $G(y) \approx \exp[-\sqrt{2 \log 2 \log(1/y)}]$  predicts  $G^{(n)}(y) \approx y^{-n} G(y)$  times powers of  $\log(1/y)$ . After  $y = e^{-v}$ , the  $L^r$  integrand has leading exponential  $\exp((nr - 1)v - r\sqrt{2Lv})$ , giving exactly the stated threshold, including convergence on the boundary  $nr = 1$ .

### 13.2 Fractional endpoint transseries

**Conjecture 13.2** (Shifted Lambert–Gamma–zeta transseries). *For fixed noninteger  $\alpha$ , the functions  $I_{0+}^\alpha F(x)$  and  $D_{0+}^\alpha F(x)$  admit full small- $x$  transseries in the same lower-Lambert coordinate as  $F$ , with the periodic Gamma–zeta phase shifted explicitly by  $\alpha$ . The leading logarithmic quadratic term is unchanged; the algebraic prefactor and periodic phase acquire computable  $\alpha$ -dependent corrections.*

*conjectural/open*

The Bromwich representations equations (93) and (94) make this a concrete saddle problem. Multiplication by  $s^{\mp\alpha}$  changes the logarithmic saddle equation by an additive  $\alpha \log s$ , while the periodic  $P(\log_2 s)$  survives.

### 13.3 Physical-space sign structure of the dyadic defect

Define for  $q > 0$

$$\Delta_q(x) = I_{0+}^q \mathcal{F}(x) - \mathfrak{A}_q \mathcal{F}(x). \quad (156)$$

Its Laplace transform is known exactly from theorem 9.6, but its zeros and sign changes in physical space are not.

**Conjecture 13.3** (Finite sign complexity on a dyadic cell). *For every  $q \in (0, 1)$ ,  $\Delta_q$  has finitely many sign changes on  $[0, 1]$ , and their number is uniformly bounded in  $q$ . The symmetry  $q \leftrightarrow 1 - q$  visible in the dominant Fourier mode of the multiplier should induce an approximate pairing of zero locations.*

*conjectural/open*

A rigorous attack could combine complete monotonicity of Laplace kernels, variation-diminishing estimates, and a certified interval evaluation of the Gamma–zeta series.

### 13.4 A Jackson or $q$ -fractional calculus at base $1/2$

The repository’s Gaussian-binomial and half-base  $q$ -Pochhammer formulas suggest that the dyadic Appell family may be only one coordinate of a richer  $q = 1/2$  fractional calculus. Promising goals are:

- construct a Jackson-type fractional integral whose integer values reproduce the exact repeated Fabius primitives;
- identify its semigroup defect as a  $q$ -cocycle related to equation (124);
- express its values at dyadic rationals by analytic continuations of Gaussian-binomial coefficients;
- compare its Mellin symbol with the Gamma–zeta multiplier  $R_q$ .

This direction may provide the most direct bridge from fractional calculus to the repository’s  $q$ -binomial arithmetic.

### 13.5 Arithmetic and closed forms for Sobolev energies

The energies  $\mathcal{E}_{2m} = \|\text{up}^{(m)}\|_2^2$  and their fractional analogues are natural periods of the infinite sinc product. It is unknown whether any nontrivial member has a closed form in familiar constants.

**Conjecture 13.4** (No elementary gamma–zeta reduction for generic energy). *Except for identities forced by scaling or Parseval, the values  $\mathcal{E}_\alpha$  at generic rational  $\alpha \geq 0$  are not finite combinations of algebraic numbers, powers of  $\pi$ , and ordinary zeta values. They should instead define a new family of dyadic sinc periods.* **conjectural/open**

Useful next steps are high-precision computation, integer-relation searches with strict complexity penalties, Mellin–Barnes representations, and comparison with the spectral zeta already developed for the zero divisor of  $\Phi$ .

### 13.6 Nonlinear fractional Volterra equations for $G$

Equation (42) can be read as a nonlinear Volterra equation because  $G$  occurs both as the integration limit and as the inverse hidden in  $F$ . Differentiating in  $y$  gives a family of Abel-type equations. Questions include uniqueness in weighted Holder classes, stable numerical inversion, and whether the order variable  $\alpha$  can be used as a regularization parameter for computing  $G$  near its singular endpoints.

### 13.7 Complex order and monodromy

The operator  $\mathfrak{A}_q$  is entire in complex  $q$  after a branch for  $2^q$  is fixed. The exact defect becomes

$$R_q(s) = \exp(P(\log_2 s + q) - P(\log_2 s)).$$

For complex  $q$ , the Fourier series of  $P$  grows in one imaginary direction and decays in the other. Determining the maximal strip of analytic continuation, the zero/pole structure of  $R_q$ , and the monodromy of complex-order fractional inversions is a concrete frontier problem.

### 13.8 Formalization program

A practical Lean development can proceed in increasing order of analytic difficulty:

1. prove the integer primitive ladder and weighted polynomial formulas from the existing global dilation API;

2. add a minimal Riemann–Liouville kernel API and prove the order-shift identity;
3. formalize the moment endpoint and fractional dyadic lattice;
4. formalize monotone substitutions for  $F$  and  $G$ , yielding the inverse transmutation theorem;
5. define  $\mathfrak{A}_q$  and prove the cocycle and derivative ladder;
6. combine the existing exact periodic decomposition with the Thue–Morse Laplace product to formalize the defect and rigidity theorem;
7. only then address Riesz kernels and the Gevrey boundary theorem.

The first five stages require mostly real integration, Gamma identities, and existing Fabius lemmas. The sixth stage reuses the strongest current Gamma–zeta formalization and would make the report’s flagship theorem machine checked.

## 14 Summary of exact formulas

For convenience, the principal identities are collected here.

| Object                        | Exact formula                                                                                                           |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| First primitive               | $\int_0^x F(t) dt = F(x/2), 0 \leq x \leq 1.$                                                                           |
| $n$ th primitive              | $I_{0+}^n F(x) = 2^{\binom{n}{2}} F(x/2^n).$                                                                            |
| $n$ th derivative             | $\mathcal{F}^{(n)}(x) = 2^{\binom{n+1}{2}} \mathcal{F}(2^n x).$                                                         |
| Up CDF                        | $\int_{-\infty}^x \text{up}(t) dt = F((x+1)/2).$                                                                        |
| Weighted fractional primitive | $I^\alpha[t^p F] = \sum_{k=0}^p (-1)^k \binom{p}{k} (\alpha)_k x^{p-k} \mathcal{J}_{\alpha+k}.$                         |
| Inverse primitive             | $\int G(y) dy = yG(y) - F(G(y)/2) + C.$                                                                                 |
| Inverse fractional integral   | $I^\alpha G(y) = \Gamma(\alpha+1)^{-1} \int_0^{G(y)} (y - F(x))^\alpha dx.$                                             |
| Inverse fractional derivative | $D^\alpha G(y) = \Gamma(1-\alpha)^{-1} \int_0^{G(y)} (y - F(x))^{-\alpha} dx, 0 < \alpha < 1.$                          |
| Mellin transform              | $\int_0^1 x^{s-1} F(x) dx = (1 - \mu(s))/s.$                                                                            |
| Fractional endpoint           | $\mathcal{J}_\alpha(1) = \mu(\alpha)/\Gamma(\alpha+1).$                                                                 |
| Fractional dyadic lattice     | $\mathcal{J}_\beta(2^{-n}) = 2^{-n\beta - \binom{n}{2}} \mu(n+\beta)/\Gamma(n+\beta+1).$                                |
| Laplace transform             | $\mathcal{L}F(s) = \phi(s)/s.$                                                                                          |
| Fourier transform of up       | $\widehat{\text{up}}(t) = \prod_{j \geq 1} \text{sinc}(t/2^j).$                                                         |
| Dyadic Appell family          | $\mathfrak{A}_q \mathcal{F}(x) = 2^{q(q-1)/2} \mathcal{F}(2^{-q} x).$                                                   |
| Exact fractional defect       | $R_q(s) = \exp(P(\log_2 s + q) - P(\log_2 s)).$                                                                         |
| Riesz tail                    | $(-\Delta)^{\alpha/2} \text{up}(x) = -C_{1,\alpha} \sum_{j \geq 0} (1+\alpha)_{2j} m_{2j} / ((2j)!  x ^{1+\alpha+2j}).$ |

| Object                  | Exact formula                                                       |
|-------------------------|---------------------------------------------------------------------|
| Inverse Mellin boundary | $\lim_{n \rightarrow \infty} (a_n / (2n)!)^{1/n} = 1 / (2 \log 2).$ |

## A Detailed proof supplements

### A.1 Entire continuation in power exponents

A recurring statement above is that an apparently restricted Mellin integral is entire. The mechanism is endpoint flatness. If  $h \in C^\infty([0, 1])$  and every derivative of  $h$  vanishes at 0, then for each  $N$  there is  $C_N$  with  $|h(x)| \leq C_N x^N$  on  $[0, 1]$ . Hence

$$\int_0^1 x^s h(x) dx$$

converges locally uniformly for  $\operatorname{Re} s > -N - 1$ . Since  $N$  is arbitrary, the Mellin transform is entire. This applies to  $F$ ,  $F'$ , and every  $F^{(n)}$  near the left endpoint. It also justifies differentiation under the integral and the analytic continuation used in equations (34) and (72).

### A.2 Moment recurrence

The distributional fixed point

$$Y \stackrel{d}{=} \frac{V + Y'}{2}, \quad (157)$$

with  $V \sim \operatorname{Unif}[0, 1]$  and  $Y' \stackrel{d}{=} Y$  independent gives, for integer  $n \geq 1$ ,

$$(2^n - 1)\mu_n = \sum_{k=0}^{n-1} \binom{n}{k} \frac{\mu_k}{n - k + 1}. \quad (158)$$

This is a convenient exact alternative to the Bell-polynomial construction. The two methods provide independent symbolic checks of the moment table.

### A.3 Why the inverse Mellin integral stops at $\operatorname{Re} s = 0$

For  $\operatorname{Re} s < 0$ ,  $F(x)^s = \exp(|\operatorname{Re} s|Q(x))$ . With  $x = e^{-u}$  and  $Q(e^{-u}) \sim cu^2$ , the integrand behaves like  $\exp(|\operatorname{Re} s|cu^2 - u)$ , which is not integrable. Thus  $A(s)$  has a genuine natural boundary for its defining integral at  $\operatorname{Re} s = 0$ . The transform  $\mathcal{M}_G(s)$  itself remains finite at  $s = 0$ , but its derivatives exhibit the same Gevrey-2 growth and it cannot cross the boundary analytically.

### A.4 A compact derivation of the Gamma–zeta defect

Writing  $u = \log s$ ,  $v = u/L$ , equation (132) is

$$\log \mathcal{L}(s) = -\frac{u^2}{2L} - \frac{u}{2} + P(v).$$

The transform of  $\mathfrak{A}_q\mathcal{F}$  is  $2^{q(q+1)/2}\mathcal{L}(2^q s)$ . Therefore

$$\begin{aligned}\log R_q(s) &= \frac{q(q+1)}{2}L + qu + \log \mathcal{L}(e^{u+qL}) - \log \mathcal{L}(e^u) \\ &= P(v+q) - P(v).\end{aligned}$$

Every polynomial term cancels. This cancellation is why the tiny periodic correction is not a secondary asymptotic detail here: it is the entire noninteger defect.

## B Numerical files and execution

The archive contains:

- `numerical_experiments.py`: the fully commented Python program;
- `numerical_results.txt`: exact moments and numerical checks;
- `moment_table.tex`: the generated rational moment table;
- `fractional_dilation_defect.png`: physical-space comparison;
- `gamma_zeta_fractional_defect.png`: transform multiplier plot.

Run from the report directory with

```
python numerical_experiments.py
```

The program requires Python 3 with `numpy`, `scipy`, `sympy`, `mpmath`, and `matplotlib`. All output paths are relative to the script.

## C Audit notes and source map

The recursive audit was organized by mathematical lineage rather than by filename count. The principal living inputs were:

- the formal-primary exposition `Fabius_Function_and_Rvachev_Up.tex`;
- the consolidated `non-formalized-research-frontiers.tex`;
- the mathematical glossary and Lean walkthrough;
- active drafts on arithmetic/dyadic sampling, inverse asymptotics, Newton–Rvachev hierarchies, half-integer spectral identities, and sinc-product Fourier decay;
- corrected/transcribed primary papers by Fabius, Rvachev, and Arias de Reyna as retained in the repository.

Searches for exact or near-exact counterparts of the new formulas used both terminology and formula fragments. In particular, no current-lineage match was found for the Riemann–Liouville order shift equation (107), the quantile transmutation equations (40) and (41), the exact defect equation (135), or the Gevrey limit equation (151). The audit does not exclude an independent result elsewhere in the mathematical literature.

## References

- [1] J. Fabius, A probabilistic example of a nowhere analytic  $C^\infty$ -function, *Zeitschrift fuer Wahrscheinlichkeitstheorie und Verwandte Gebiete* **5** (1966), 173–174.
- [2] V. L. Rvachev, On a finite function, *Doklady Akademii Nauk Ukrainian SSR*, Series A (1971), no. 8, 705–707.
- [3] V. L. Rvachev, Compactly supported solutions of functional-differential equations and their applications, *Russian Mathematical Surveys* **45** (1990), no. 1, 87–120.
- [4] J. Arias de Reyna, On the Fabius function, original Spanish article (1982); English translation and source copy retained in the project documentation.
- [5] J. Arias de Reyna, Arithmetic of the Fabius function, *INTEGERS* **18** (2018), Paper A48; arXiv:1702.06487.
- [6] J.-P. Allouche and J. Shallit, *Automatic Sequences: Theory, Applications, Generalizations*, Cambridge University Press, 2003.
- [7] S. G. Samko, A. A. Kilbas, and O. I. Marichev, *Fractional Integrals and Derivatives: Theory and Applications*, Gordon and Breach, 1993.
- [8] M. Caputo, Linear models of dissipation whose  $Q$  is almost frequency independent–II, *Geophysical Journal of the Royal Astronomical Society* **13** (1967), 529–539.
- [9] M. Kwasnicki, Ten equivalent definitions of the fractional Laplace operator, *Fractional Calculus and Applied Analysis* **20** (2017), 7–51; arXiv:1507.07356.
- [10] N. D. Cong, Semigroup property of fractional differential operators and its applications, arXiv:2107.08914, 2021.
- [11] R. M. Corless, G. H. Gonnet, D. E. G. Hare, D. J. Jeffrey, and D. E. Knuth, On the Lambert  $W$  function, *Advances in Computational Mathematics* **5** (1996), 329–359.
- [12] V. Reshetnikov et al., *The Fabius Function and Rvachev Up: definitions, equivalences, arithmetic, asymptotics, and formalization map*, living LaTeX exposition in the ProveIt repository, accessed 27 August 2026.
- [13] V. Reshetnikov et al., *Non-formalized research frontiers for the Fabius and Rvachev functions*, consolidated LaTeX frontier volume in the ProveIt repository, accessed 27 August 2026.